

Stability and Generalization in Learning Theory

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Outline

- 1 Background
- 2 Uniform Stability
- 3 On-average Model Stability
- 4 Conclusion

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Population and Empirical Risk

Let P be a probability distribution defined on a sample space $\mathcal{Z} := \mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} \subseteq \mathbb{R}^d$ and $\mathcal{Y} \subseteq \mathbb{R}$. Let $S = \{z_i = (x_i, y_i) : i = 1, \dots, n\}$ be sampled drawn from P .

We wish to build a model $f : \mathcal{X} \rightarrow \mathbb{R}$. Suppose f is parameterized by w . The performance of the model is usually measured by a loss function ℓ :

- Regression task: $\ell(w; z) = \frac{1}{2}(y - f(w; x))^2$
- Classification task: $\ell(w; z) = \log(1 + \exp(-yf(w; x)))$

which is used to get population risk and empirical risk

$$L_S(w) = \frac{1}{n} \sum_{i=1}^n \ell(w; z_i), \quad L(w) = \mathbb{E}_{z \sim P}[\ell(w; z)]$$

The ability of the trained model to perform well is called generalization, which is usually measure by so-called generalization error, which is crucial in both practice and theory:

Generalization Error

$$\epsilon_{gen} = \mathbb{E}_S(L(w) - L_S(w))$$

Optimization Algorithm

We want to get a model w with a small population risk.

This is achieved by an optimization algorithm A , which could output $A(S)$!

- empirical risk minimization (ERM): $A(S) = \arg \min_w L_S(w)$
- regularized risk minimization: $A(S) = \arg \min_w \{L_S(w) + \lambda \|w\|_2^2\}$
- gradient descent, stochastic gradient descent

Gradient Descent(GD)

Initialize $w = w_0$, and at iteration t , perform

$$w_t = w_{t-1} - \eta_t \nabla_w L_S(w_{t-1})$$

Schochastic Gradient Descent(SGD)

Initialize $w = w_0$, and at iteration t , randomly choosing $i_t \in [1, \dots, n]$, then perform

$$w_t = w_{t-1} - \eta_t \nabla_w \ell(w_{t-1}; z_{i_t})$$

Existing Work on Generalization Analysis

Recently, overparameterized networks trained by first-order methods have show great success practically, but theoretical analysis is still far behind. In this lecture, we mainly focus on generalization error.

- uniform convergence: VC dimension, Rademacher complexities, covering numbers (capacity-based bounds)
- NTK approach (high parameterization)

However, these methods do not take account for algorithms or the overparameterization requirement is too high!

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Uniform Stability Approach

Uniform Stability

A stochastic algorithm A is ϵ -uniformly stable if for all training datasets $S, \tilde{S} \in \mathcal{Z}^n$ that differ by at most one example, we have

$$\sup_z \mathbb{E}_A \left[\ell(A(S); z) - \ell(A(\tilde{S}); z) \right] \leq \epsilon.$$

Theorem 1 (Generalization via uniform stability)

Let A be ϵ -uniformly stable. Then

$$|\mathbb{E}_{S,A} [L_S(A(S)) - L(A(S))]| \leq \epsilon.$$

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Let $S = \{z_1, \dots, z_n\}$ and $S' = \{z'_1, \dots, z'_n\}$ be drawn independently from P . For any $i \in [n]$, define $S^{(i)} = \{z_1, \dots, z_{i-1}, z'_i, z_{i+1}, \dots, z_n\}$ as the set formed from S by replacing the i -th element with z'_i .

Uniform Stability Approach

McDiarmid

Let $F : \mathcal{Z}^n \rightarrow \mathbb{R}$ be any measurable function for which there exists constants $c_i, 1 \leq i \leq n$ such that

$$\sup |F(S) - F(S^{(i)})| \leq c_i,$$

then

$$P_S[F(S) - \mathbb{E}_S[F(S)] \leq \epsilon] \leq e^{-2\epsilon^2 / \sum_{i=1}^n c_i^2}$$

with the concentration inequality, we can get high-probability bounds

Theorem 2

Assume that the loss function ℓ is bounded by $M \geq 0$. A stochastic algorithm A is ϵ -uniformly stable. Then with probability with at least $1 - \delta$ over the sampling, the following holds:

$$L_S(A(S)) \leq L(A(S)) + \epsilon + (2n\epsilon + M) \sqrt{\frac{\log(1/\delta)}{2n}}$$

There are also other forms of stability, hypothesis stability, pointwise hypothesis stability, error stability, etc.

Properties of Update Rules

For an update rule $G(w) = w - \eta \nabla \ell(w)$, we define the following properties

Expansiveness

An update rule is α -expansive if

$$\sup_{v, w \in \Omega} \frac{\|G(v) - G(w)\|}{\|v - w\|} \leq \alpha.$$

σ -boundedness

An update rule is σ -bounded if

$$\sup_{w \in \Omega} \|w - G(w)\| \leq \sigma.$$

Now we control how a sequence of updates to a model diverge when the training set is perturbed.

Growth Recursion

Lemma 3

Fix an arbitrary sequence of updates $G_1, \dots, G_T$ and another sequence $G'_1, \dots, G'_T$. Let $w_0 = w'_0$ be a starting point in Ω and define $\delta_t = \|w'_t - w_t\|$ where w_t, w'_t defined recursively through

$$w_{t+1} = G_t(w_t) \quad w'_{t+1} = G'_t(w'_t).$$

Then, we have the recurrence relation

$$\delta_0 = 0$$
$$\delta_{t+1} \leq \begin{cases} \alpha \delta_t & G_t = G'_t \text{ is } \alpha\text{-expansive} \\ \min(\alpha, 1) \delta_t + 2\sigma_t & G_t \text{ and } G'_t \text{ are } \sigma\text{-bounded,} \\ G_t \text{ is } \alpha \text{ expansive} & (t > 0). \end{cases}$$

Expansiveness of Stochastic Gradient

Different conditions of ℓ may lead to different expansive properties.

Definition 4

A function $\ell : \Omega \rightarrow \mathbb{R}$ is β -smooth if for all $u, v \in \Omega$ we have

$$\|\nabla \ell(u) - \nabla \ell(v)\| \leq \beta \|u - v\|.$$

Now we assume ℓ is L -Lipshitz and β -smooth, then the gradient update rule has following properties:

- G is ηL - bounded.
- G is $1 + \eta\beta$ -expansive.
- Assume in addition that ℓ is convex, then for any $\eta \leq 2/\beta$, G is 1-expansive.
- Assume in addition that ℓ is γ -strongly convex, then for any $\eta \leq 2/\beta$, G is $1 - \frac{\eta\beta\gamma}{\beta+\gamma}$ -expansive.

Uniform Stability of SGD

In convex cases

Theorem 5

Assume that the loss function $\ell(\cdot; z)$ is β -smooth, convex and L -Lipschitz for every z . Suppose that we run SGD with step sizes $\eta_t \leq 2/\beta$ for T steps. Then, SGD satisfies uniform stability with

$$\epsilon_{stab} \leq \frac{2L^2}{n} \sum_{t=1}^T \eta_t.$$

Uniform Stability of SGD

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In strongly convex cases, we use PSGD, i.e. $w_{t+1} = \Pi_{\Omega}(w_t - \eta_t \nabla \ell(w_t; z_t))$.

Theorem 6

Assume that the loss function $\ell(\cdot; z)$ is β -smooth, γ -strongly convex and L -Lipschitz for every z . Suppose that we run PSGD with step sizes $\eta_t \leq 1/\beta$ for T steps. Then, SGD satisfies uniform stability with

$$\epsilon_{stab} \leq \frac{2L^2}{\gamma n}$$

Nonconvex Cases

Lemma 7

Assume that the loss function $\ell(\cdot; z)$ is nonnegative and L -Lipschitz for all z . Let S and S' be two samples of size n differing in only a single example. Denote by w_T and w'_T the output of T steps of SGD on S and S' , respectively. Then, for every $z \in Z$ and every $t_0 \in \{0, 1, \dots, n\}$, under both the random update rule and the random permutation rule, we have

$$\mathbb{E}|\ell(w_T; z) - \ell(w'_T; z)| \leq \frac{t_0}{n} \sup_{w, z} \ell(w; z) + L\mathbb{E}[\delta_T \mid \delta_{t_0} = 0]$$

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Using similar proof techniques in convex optimization, we get

Theorem 8

Assume that $\ell(\cdot; z) \in [0, 1]$ is an L -Lipschitz and β -smooth loss function for every z . Suppose that we run SGD for T steps with monotonically non-increasing step sizes $\eta_t \leq c/t$. Then, SGM has uniform stability with

$$\epsilon_{stab} \leq \frac{1 + 1/\beta c}{n - 1} (2cL^2)^{\frac{1}{\beta c + 1}} T^{\frac{\beta c}{\beta c + 1}} \lesssim \frac{T^{1 - 1/(\beta c + 1)}}{n}$$

Stability-inducing operations

- weight decay and regularization: expansiveness
- gradient clipping: Lipschitzness
- dropout: $DG(v) = v - \eta D(\nabla \ell(v))$ Lipschitzness
- projections and proximal steps

Definition 9

For a nonnegative step size $\alpha \geq 0$ and a function $f : \Omega \rightarrow \mathbb{R}$, we define the proximal update rule $P_{f,\alpha}$ as

$$P_{f,\alpha}(w) = \arg \min_v \left\{ \frac{1}{2} \|w - v\|^2 + \alpha f(v) \right\}.$$

Then if f is convex, the proximal update is 1-expansive.

- model averaging: same assumptions as Theorem (5), and $\eta_t \leq \eta$, then $\epsilon_{\text{stab}} \leq \frac{\eta TL^2}{n}$.

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- 1 Background
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Fine-grained analysis

Above methods require restrictive assumptions on the boundedness of gradients, smoothness and convexity of loss functions.

- Nonsmoothness: ReLU networks, q -norm loss
 $\ell(w; z) = |y - \langle w, x \rangle|^q (q \in [1, 2])$ and hinge loss $(1 - y\langle w, x \rangle)_+$ for $x \in \mathbb{R}^d$
- Nonconvexity: networks
- Nonlipshithness: unbounded domain for quadratic loss

We need fine-grained analysis of stability and generalization for SGD!

On-average Model Stability

The key to remove the bounded gradient assumption and the strong smoothness assumption is the introduction of a novel stability measure.

Definition 10

Let $S = \{z_1, \dots, z_n\}$ and $S' = \{z'_1, \dots, z'_n\}$ be drawn independently from P . For any $i \in [n]$, define $S^{(i)} = \{z_1, \dots, z_{i-1}, z'_i, z_{i+1}, \dots, z_n\}$ as the set formed from S by replacing the i -th element with z'_i . We say a randomized algorithm A is ℓ_1 -on-average model ϵ -stable if

$$\mathbb{E}_{S, S', A} \left[\frac{1}{n} \sum_{i=1}^n \|A(S) - A(S^{(i)})\|_2 \right] \leq \epsilon.$$

and ℓ_2 -on-average model ϵ -stable if

$$\mathbb{E}_{S, S', A} \left[\frac{1}{n} \sum_{i=1}^n \|A(S) - A(S^{(i)})\|_2^2 \right] \leq \epsilon^2.$$

Generalization via Model Stability

The connection between the generalization error and the on-average model stability was established in the following lemmas.

Lemma 11

Let A be ℓ_1 -on-average model ϵ -stable and $\ell(\cdot; z)$ is L -Lipschitz, then

$$\mathbb{E}[L(A(S)) - L_S(A(S))] \leq L\epsilon.$$

Lemma 12

Let A be a stochastic algorithm. If for any z , the map $W \mapsto \ell(W; z)$ is ρ -smooth and nonnegative, then

$$\begin{aligned} \mathbb{E}[L(A(S)) - L_S(A(S))] &\leq \frac{\rho}{2n} \sum_{i=1}^n \mathbb{E}[\|A(S) - A(S^{(i)})\|_2^2] \\ &\quad + \left(\frac{2\rho \mathbb{E}[L_S(A(S))]}{n} \sum_{i=1}^n \mathbb{E}[\|A(S) - A(S^{(i)})\|_2^2] \right)^{\frac{1}{2}}. \end{aligned}$$

Generalization via Model Stability

The following is an example for nonsmooth function.

Definition 13

Let $L > 0$, $\alpha \in [0, 1]$. We say $\partial\ell$ is (α, L) -Hölder continuous if for all $w, \tilde{w} \in \mathbb{R}^d$ and $z \in Z$,

$$\|\partial\ell(w, z) - \partial\ell(\tilde{w}, z)\|_2 \leq L \|w - \tilde{w}\|_2^\alpha.$$

Lemma 14

If for any z , the function $w \mapsto \ell(w; z)$ is nonnegative, convex and $w \mapsto \partial\ell(w; z)$ is (α, L) -Hölder continuous with $\alpha \in [0, 1]$, then

$$\begin{aligned} \mathbb{E}_{S,A} [L(A(S)) - L_S(A(S))] &\leq \frac{c_{\alpha,1}^2}{2\gamma} \mathbb{E}_{S,A} \left[L^{\frac{2\alpha}{1+\alpha}}(A(S)) \right] \\ &\quad + \frac{\gamma}{2n} \sum_{i=1}^n \mathbb{E}_{S,\tilde{S},A} \left[\left\| A(S^{(i)}) - A(S) \right\|_2^2 \right]. \end{aligned}$$

Stability and Generalization for Neural Networks

We focus on the stability analysis for two-layer neural networks with smooth activations, which is nonconvex.

$$f(W; x) := \sum_{k=1}^m \mu_k \sigma(\langle w_k, x \rangle),$$

where we fix $\mu_k \in \left\{ \frac{1}{\sqrt{m}}, -\frac{1}{\sqrt{m}} \right\}$, $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is an activation function, and $W = (w_1, \dots, w_m) \in \mathbb{R}^{d \times m}$ is the weight matrix.

Assumption 1 (Activation)

The activation $\phi(u)$ is continuous and twice differentiable with constant $B_\phi, B_{\phi'}, B_{\phi''} > 0$ bounding $|\phi(u)| \leq B_\phi$, $|\phi'(u)| \leq B_{\phi'}$, and $|\phi''(u)| \leq B_{\phi''}$ for any $u \in \mathbb{R}$.

Assumption 2 (Inputs, labels, and the loss function)

There exists constants $C_x, C_y, C_0 > 0$ such that $\|\mathbf{x}\|_2 \leq C_x$, $|y| \leq C_y$ and $\ell(\mathbf{W}_0; \mathbf{z}) \leq C_0$ for any $\mathbf{x}, y$ and $\mathbf{z}$.

Regression Task

We consider MSE loss: $\ell(W; z) = \frac{1}{2}(y - f(W; x))^2$

The following lemma shows that the loss function is smooth and the loss function is weakly convex. We develop a lower bound for the eigenvalue of the Hessian matrix. Let $\lambda_{\min}(A)$ denote the smallest eigenvalue of a matrix A and $\nabla^2 f$ denote the Hessian matrix of a function f . We use $a \vee b = \max\{a, b\}$ for any $a, b \in \mathbb{R}$.

Technical Lemmas

Lemma 15 (Smoothness and Curvature)

Let $\mathbf{z} \in \mathcal{Z}$. The function $\mathbf{W} \mapsto \ell(\mathbf{W}; \mathbf{z})$ is ρ -smooth. For any $\mathbf{W}$, we have

$$\lambda_{\min}(\nabla^2 \ell(\mathbf{W}; \mathbf{z})) \geq -\frac{b'}{\sqrt{m}} (\|\mathbf{W} - \mathbf{W}_0\|_2 \vee 1).$$

Technical Lemmas

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$$\lambda_{\min}(\nabla^2 \ell(\mathbf{W}; \mathbf{z})) \geq -\frac{b'}{\sqrt{m}} (\|\mathbf{W} - \mathbf{W}_0\|_2 \vee 1).$$

Lemma 16 (Taylor's expansion)

Let $\mathbf{W}, \mathbf{W}' \in \mathbb{R}^{d \times m}$. Then

$$\ell(\mathbf{W}; \mathbf{z}) - \ell(\mathbf{W}'; \mathbf{z}) - \langle \mathbf{W} - \mathbf{W}', \nabla \ell(\mathbf{W}'; \mathbf{z}) \rangle \geq -\frac{b'R}{\sqrt{m}} \|\mathbf{W} - \mathbf{W}'\|_2^2,$$

where $R = \max\{1, \|\mathbf{W} - \mathbf{W}_0\|_2, \|\mathbf{W}' - \mathbf{W}_0\|_2\}$.

Technical Lemmas

Lemma 17 (Self-bounding property)

Assume for all $\mathbf{z}$, the function $\mathbf{w} \mapsto \ell(\mathbf{w}; \mathbf{z})$ is nonnegative and ρ -smooth. Then

$$\|\nabla \ell(\mathbf{w}; \mathbf{z})\|_2^2 \leq 2\rho \ell(\mathbf{w}; \mathbf{z}).$$

Technical Lemmas

Lemma 17 (Self-bounding property)

Assume for all $\mathbf{z}$, the function $\mathbf{w} \mapsto \ell(\mathbf{w}; \mathbf{z})$ is nonnegative and ρ -smooth. Then

$$\|\nabla \ell(\mathbf{w}; \mathbf{z})\|_2^2 \leq 2\rho \ell(\mathbf{w}; \mathbf{z}).$$

Lemma 18 (Deviation from the initial point)

Let Assumptions 1, 2 hold and assume $\eta \leq 1/(2\rho)$. Let $\{\mathbf{W}_t\}$ be produced by gradient descent. Then for any $t \in \mathbb{N}$ we have

$$\|\mathbf{W}_t - \mathbf{W}_0\|_2 \leq \sqrt{2\eta t L_S(\mathbf{W}_0)}.$$

Almost co-coercivity of the Gradient Operator

Lemma 19

Let Assumptions 1, 2 hold. If $\eta \leq 1/(2\rho)$, then for any $t \in \mathbb{N}$ we have

$$\begin{aligned} & \langle \mathbf{W}_t - \mathbf{W}_t^{(i)}, \nabla \ell(\mathbf{W}_t; \mathbf{z}_i) - \nabla \ell(\mathbf{W}_t^{(i)}; \mathbf{z}_i) \rangle \\ & \geq 2\eta \left(1 - \frac{\eta\rho}{2}\right) \|\nabla \ell(\mathbf{W}_t; \mathbf{z}_i) - \nabla \ell(\mathbf{W}_t^{(i)}; \mathbf{z}_i)\|_2^2 \\ & \quad - \epsilon'_t \left\| \mathbf{W}_t - \mathbf{W}_t^{(i)} - \eta(\nabla \ell(\mathbf{W}_t; \mathbf{z}_i) - \nabla \ell(\mathbf{W}_t^{(i)}; \mathbf{z}_i)) \right\|_2^2, \end{aligned}$$

where

$$\epsilon'_t = \frac{C_x^2 B_{\phi''}}{\sqrt{m}} \left(B_{\phi'} C_x (1 + 2\eta\rho) \max\{\|\mathbf{W}_t - \mathbf{W}_0\|_2, \|\mathbf{W}_t^{(i)} - \mathbf{W}_0\|_2\} + \sqrt{2C_0} \right).$$

Iterates of stability

Lemma 20

Let Assumptions 1, 2 hold. If $\eta \leq 1/(2\rho)$, then for any $t \in \mathbb{N}$ we have

$$\begin{aligned} \|\mathbf{W}_{t+1} - \mathbf{W}_{t+1}^{(i)}\|_2^2 &\leq \frac{1 + \rho}{1 - 2\eta\epsilon_t} \|\mathbf{W}_t - \mathbf{W}_t^{(i)}\|_2^2 \\ &\quad + \frac{2(1 + 1/\rho)\eta^2}{n^2} \left(\|\nabla \ell(\mathbf{W}_t; \mathbf{z}_i)\|_2^2 + \|\nabla \ell(\mathbf{W}_t^{(i)}; \mathbf{z}'_i)\|_2^2 \right), \end{aligned}$$

where

$$\epsilon_t = \frac{C_x^2 B_{\phi''}}{\sqrt{m}} \left(B_{\phi'} C_x (1 + \eta\rho) \|\mathbf{W}_t - \mathbf{W}_t^{(i)}\|_2 + 2\sqrt{2C_0} \right).$$

From which we can get

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E} \|\mathbf{W}_{t+1} - \mathbf{W}^{(i)}\|_2^2 \leq \frac{\eta^2 t}{n^2} 16e^3 C_x^2 B_{\phi'}^2 \sum_{j=0}^n \mathbb{E} L_S(\mathbf{W}_j)$$

Generalization Error

Theorem 21

Consider Assumptions 1 and 2. Fix $t > 0$. If $\eta \leq 1/(2\rho)$ and

$$m \geq 144(\eta t)^2 C_x^4 C_0^2 B_{\phi''}^2 \left(4B_{\phi'} C_x \sqrt{\eta t} + \sqrt{2} \right)^2$$

then

$$\epsilon_{gen}(\mathbf{W}_{t+1}) \leq b \left(\frac{\eta}{n} + \frac{\eta^2 t}{n^2} \right) \sum_{j=0}^t \mathbb{E} L_S(W_j)$$

where

$$b = 16e^3 C_x^{\frac{3}{2}} B_{\phi'}^2 \left(1 + C_x^{\frac{3}{2}} B_{\phi'}^2 \right).$$

Optimization Error

Theorem 22

Consider Assumptions 1 and 2. Fix $t > 0$. If $\eta \leq 1/(2\rho)$, then

$$\frac{1}{t} \sum_{j=0}^t L_S(\mathbf{W}_j) \leq \min_{\mathbf{W} \in \mathbb{R}^{d \times m}} \left\{ L_S(\mathbf{W}) + \frac{\|\mathbf{W} - \mathbf{W}_0\|_F^2}{\eta t} + \frac{\tilde{b} \|\mathbf{W} - \mathbf{W}_0\|_F^3}{\sqrt{m}} \right\} + \tilde{b} C_0 \cdot \frac{(\eta t)^{\frac{3}{2}}}{\sqrt{m}}$$

where $\tilde{b} = C_x^2 B_{\phi''} (B_{\phi'} C_x + C_0)$.

which is related to the following regularised empirical risk minimization:

$$\min_{W \in \mathbb{R}^{d \times m}} L_S(W) + \lambda \|W - W_1\|_F^2$$

Estimation of ERM

- NTK solution, under some wild assumptions, ERM term is close to $\frac{1}{n} \langle \mathbf{y}, (n\mathbf{K})^{-1} \mathbf{y} \rangle$
- The following regularity assumption is standard in learning theory:

$$\mathbf{W}_\lambda^* = \arg \min_{\mathbf{W} \in \mathbb{R}^{d \times m}} \{L(\mathbf{W}) + \lambda \|\mathbf{W} - \mathbf{W}_0\|_2^2\}.$$

Assumption 3 (Regularity)

Assume there exist $\alpha \in (0, 1]$ and $c_\alpha > 0$ such that

$$\Lambda_\lambda := L(\mathbf{W}_\lambda^*) - L(\mathbf{W}^*) + \lambda \|\mathbf{W}_\lambda^* - \mathbf{W}_0\|_2^2 \leq c_\alpha \lambda^\alpha.$$

Classification Task

Use the same two-layer neural network for classification, with logistic loss:

$$\ell(W; z) = \log(1 + \exp(-yf(W; x))).$$

Logistic loss has better self-bounding properties than quadratic loss:

- Self-bounded gradient: $\|\nabla \ell(W; z_i)\| \leq B_{\phi'} C_x \ell(W; z_i), i \in [n]$.
- Curvature bound: $\lambda_{\min}(\nabla^2 L_S(W)) \geq -\frac{B_{\phi''} C_x^2}{\sqrt{m}} L_S(W)$.

Generalized Local Quasi-Convexity

Taylor's theorem implies:

$$L_S(W) \geq L_S(W_t) + \langle \nabla L_S(W_t), W - W_t \rangle + \frac{1}{2} \lambda_{\min}(\nabla^2 L_S(W_{\alpha t})) \|W - W_t\|^2.$$

By self-bounding property, $L_S(W_{\alpha t})$ is controlled via:

Lemma 23

For any W_1, W_2 , $\lambda > 1$, and $\sqrt{m} \geq \frac{\lambda B_{\phi''} C_x^2}{2} \|W_1 - W_2\|^2$:

$$\max_{V \in [W_1, W_2]} L_S(V) \leq (1 - 1/\lambda)^{-1} \cdot \max\{L_S(W_1), L_S(W_2)\}.$$

Training Loss

Theorem 24

If $\|W - W_0\|^2 \geq \max\{\eta TL_S(W), \eta L_S(W_0)\}$ and $m \geq 18^2 B_{\phi''}^2 C_x^4 \|W - W_0\|^4$, then:

$$L_S(W_T) \leq \frac{1}{T} \sum_{t=1}^T L_S(W_t) \leq 2L_S(W) + \frac{5\|W - W_0\|^2}{2\eta T},$$

$$\forall t \in [T] : \|W_t - W_0\| \leq 4\|W - W_0\|.$$

Here, the ERM can be controlled under margin assumptions.

Generalization Error

Lemma 25 (Expansiveness for self-bounded losses)

Provided $\eta \leq 1/(B_\phi^2, C_x^2)$, it holds that

$$\begin{aligned} & \| (W - \eta \nabla L_S(W)) - (W' - \eta \nabla L_S(W')) \| \\ & \leq \left(1 + \eta \frac{B_{\phi''} C_x^2}{\sqrt{m}} \max_{\alpha \in [0,1]} L_S(W_\alpha) \right) \| W - W' \|. \end{aligned}$$

Therefore, we get

Theorem 26

For any step size $\eta \leq 1/\rho$, let any W such that

$\|W - W_0\|^2 \geq \max\{\eta TL_S(W), \eta L_S(W_0)\}$. Suppose hidden-layer width m satisfies $m \geq 64^2 B_{\phi''}^2 C_x^4 \|W - W_0\|^4$. Then, the generalization gap of GD at iteration T is bounded as

$$\mathbb{E}[L(W_T) - L_S(W_T)] \leq \frac{8B_{\phi'}^2 C_x^2}{n} \mathbb{E}[\eta TL_S(W) + 2\|W - W_0\|^2]$$

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- 1 Background
- 2 Uniform Stability
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Conclusion

Conclusion

- Algorithmic stability is an effective approach to estimating generalization error in machine learning theory, e.g. SGD in convex and smooth problems.
- Introduction to on-average model stability and how to use it to analyze non-convex functions like shallow neural networks in both regression and classification problems.

Future work

- How to apply it to ReLU networks despite of its nonsmooth activation?
- How to control the norm of reference model?
- How to apply algorithmic stability to study generalization in other fields, such as GNN, adversarial learning, contrastive learning, large language model, etc.